Lattice depletion, signal time advances, and effective repulsion

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We consider a lattice modification in which intermediate processing nodes are removed and the remaining nodes are reconnected, so that a signal crossing a region encounters fewer elementary delays than in a reference lattice. Under stated continuum and isotropy assumptions, the relative encounter count becomes an optical index n. Depletion, $n < 1$, produces a travel-time advance; a radially monotone central depletion deflects passing rays outward. These results concern optical geometry and do not require assigning negative energy to missing constituents. A gravitational interpretation additionally requires a prescription for clocks and massive probes. With the usual static weak-field metric, the same depletion corresponds to a positive potential and repulsive free fall. A smooth example makes the distinction explicit, while a negative-mass type exterior requires a $1/r$ encounter deficit. The model formalizes the intuition that fewer processing steps reverse the optical effect of additional delays, and identifies the further assumptions needed to interpret that reversal as gravity.

I. THE PROPOSED OPERATION

Suppose that signal propagation through a microscopic substrate takes time because a signal undergoes successive local processing events. Adding events along a route then increases its traversal time. The question considered here is the converse: what would observers built from the substrate experience if a traversable route required fewer events?

The intended operation is *depletion with reconnection*. A sequence of processing nodes is shortened while passage between its surviving endpoints is retained. The delay associated with a removed node is omitted; it is not reassigned to the new connection. Thus a region with fewer nodes can be crossed earlier than the corresponding region of an undepleted reference lattice. This propagation rule is the defining physical assumption of the model.

Geometric descriptions of lattice defects provide a broader setting for this question [1]. For embedded observers we follow Toffoli's discussion of observers within uniform systems [2]. Here the observer and its measuring devices belong to the same substrate. The processing rule specified here is an additional assumption. We derive its optical consequences before introducing a gravitational interpretation. The distinction between a propagation metric and gravitational dynamics is also familiar from acoustic analogues [3].

The reference lattice supplies the comparison meant by "fewer." Its coordinates label corresponding regions before and after modification; they need not coincide with the distances inferred inside the modified system. We take the depletion to be static and the exterior to provide an ordinary reference region.

II. FROM PROCESSING EVENTS TO AN OPTICAL METRIC

A. Discrete rule and continuum assumptions

Let a be the microscopic length scale of a family of reference networks. If a route γ_a contains $N_a[\gamma_a]$ forwarding events, its traversal time is

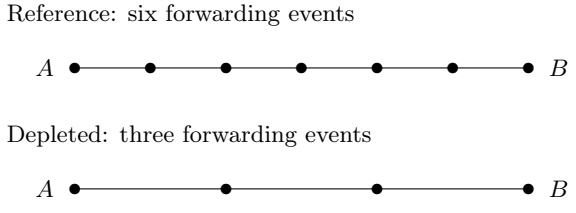


FIG. 1. The elementary operation. Each traversal to the next retained node incurs one delay τ_a. Removing intermediate nodes and reconnecting the sequence reduces the event count at fixed reference endpoints. The drawing represents the microscopic rule; the continuum model additionally requires local reconnections and isotropic propagation.

$$T_a[\gamma_a] = \tau_a N_a[\gamma_a], \qquad \tau_a = \frac{a}{c_0}. \tag{1}$$

Within each network every event has the same elementary delay τ_a. Its scaling with a keeps the reference signal speed c_0 finite in a continuum limit. Counting or omitting a fixed number of endpoint events does not affect that limit. A finite nonzero continuum advance therefore requires the number of omitted events along a macroscopic route to be of order $1/a$; removing only finitely many events has a vanishing effect.

For example, replacing six forwarding events by three gives $T_a = 3\tau_a$ instead of $6\tau_a$ (Fig. 1). More generally, for corresponding routes with $N_a = (1-\varepsilon)N_{0,a}$,

$$T_a = (1-\varepsilon)T_{0,a}, \qquad \Delta T_a = -\varepsilon T_{0,a}. \tag{2}$$

This is the elementary time advance to be retained after coarse graining.

We make the following continuum assumptions. Reconnections are local: their extent in reference coordinates tends to zero with a. On scales large compared with a but small compared with the variation scale of the depletion, the encounter count is isotropic. For suitable microscopic approximations γ_a of a smooth curve γ,

we assume

$$aN_a[\gamma_a] \longrightarrow \int_\gamma n(\mathbf{x})\,\mathrm{d}\ell, \qquad \mathrm{d}\ell = |\mathrm{d}\mathbf{x}|. \tag{3}$$

We assume that minimum travel times converge to the minimum of this functional as well. These are assumptions about the family of networks, not consequences of deletion alone. For example, an axis-aligned nearest-neighbor cubic network retains a directional shortest-path norm when refined; refinement alone does not establish the isotropy used in Eq. (3).

The dimensionless field n is an *encounter density along a path*, relative to the reference value. It is not, in general, the occupied fraction of nodes per unit volume. We require n to be smooth on the continuum scale and bounded away from zero, with $n \to 1$ in the reference exterior. A depleted region has $n < 1$.

B. Travel times

Equations (1) and (3) give

$$\begin{aligned} T[\gamma] &= \frac{1}{c_0}\int_\gamma n\,\mathrm{d}\ell, \\ T_n(A,B) &= \frac{1}{c_0}\inf_{\gamma:A\to B}\int_\gamma n\,\mathrm{d}\ell. \end{aligned} \tag{4}$$

The optical metric is therefore

$$\mathrm{d}\ell_{\mathrm{opt}}^2 = n^2(\mathbf{x})\,\mathrm{d}\mathbf{x}^2. \tag{5}$$

It encodes distances measured by reference travel time, $\mathrm{d}\ell_{\mathrm{opt}} = c_0\mathrm{d}t$ along a ray.

There is a simple monotonicity statement. For the same endpoints and admissible continuum paths, $n_1 \le n_2$ implies

$$T_{n_1}(A,B) \le T_{n_2}(A,B). \tag{6}$$

The inequality holds for each path integral and hence for their infima. Thus $n \le 1$ gives a time advance or equality relative to the reference. If the straight reference segment is admissible and passes through $n < 1$ for a nonzero length, then $T_n(A,B) < |B-A|/c_0$.

For a weak depletion

$$n(\mathbf{x}) = 1 - \varepsilon f(\mathbf{x}), \qquad 0 \le f \le 1, \quad 0 < \varepsilon \ll 1, \tag{7}$$

the first-order change between fixed endpoints is

$$\Delta T = -\frac{\varepsilon}{c_0}\int_{\gamma_0} f\,\mathrm{d}\ell + O(\varepsilon^2), \tag{8}$$

where γ_0 is the unique straight reference path. Its stationarity makes the first-order displacement of the path irrelevant to this expression. The remainder is understood at fixed endpoint geometry.

III. RAYS AND INTRINSIC MEASUREMENTS

Varying the travel-time functional, with $\mathbf{e} = \mathrm{d}\mathbf{x}/\mathrm{d}\ell$ the Euclidean unit tangent, gives

$$\frac{\mathrm{d}}{\mathrm{d}\ell}(n\mathbf{e}) = \nabla n, \qquad \frac{\mathrm{d}\mathbf{e}}{\mathrm{d}\ell} = (\mathbf{I} - \mathbf{e}\mathbf{e}^{\mathsf{T}})\nabla \ln n. \tag{9}$$

Rays bend toward increasing encounter density. For a radially depleted centre, $n(r)$ increases outward, so a passing ray bends away from the centre. A weak localized depletion therefore acts as a diverging lens. Additional central delays, for which n decreases outward toward its reference value, give the opposite deflection. Both conclusions follow from the same processing rule.

An intrinsic observer need not describe the effect as a locally higher signal speed. If distance is reconstructed using signal exchanges, fewer encountered nodes mean a shorter optical distance. Observable comparisons can use clocks and endpoints in the ordinary exterior, or compare alternative routes with different exposure to the depleted region. They establish relative travel times and ray geometry without directly counting microscopic nodes. Inferring depletion as their microscopic cause is then a model interpretation.

A uniform change that preserves all dimensionless internal relationships can be absorbed into a choice of units. Likewise, in one spatial dimension the optical line element becomes $\mathrm{d}X^2$ under

$$X(x) = \int^x n(u)\,\mathrm{d}u. \tag{10}$$

Its variable node count alone does not establish intrinsic curvature. In two or more spatial dimensions a position-dependent isotropic optical metric is generally curved. Curvature depends on derivatives of n, however; the inequality $n < 1$ is not itself a statement that curvature is negative everywhere.

IV. WHAT IS REQUIRED FOR A GRAVITATIONAL INTERPRETATION?

A. The clock prescription

Optical data do not select a unique spacetime metric. Consider the static family

$$\mathrm{d}s^2 = \Omega^2(\mathbf{x})\left[-c_0^2\mathrm{d}t^2 + n^2(\mathbf{x})\mathrm{d}\mathbf{x}^2\right], \tag{11}$$

with $\Omega > 0$ and $\Omega \to 1$ in the reference exterior. Every member has the travel-time law (4) and the same unparameterized null rays. Stationary ideal clocks, in contrast, measure

$$\mathrm{d}\tau = \Omega\,\mathrm{d}t. \tag{12}$$

This conformal freedom is the usual distinction between ray geometry and clock-sensitive spacetime geometry [3].

The choice $\Omega = 1$ is consistent with identical clocks built from a fixed number of equal-delay processing events. It gives the optical effect but no clock-rate gradient; initially stationary massive geodesics remain stationary. The choice $\Omega = 1/n$ has the same optics and position-dependent clock rates. Thus a universal clock and probe dynamics must be supplied before optical defocusing can be identified with repulsive free fall. Merely stipulating equal microscopic delays does not derive that dynamics.

B. A weak-field completion

To compare the two optical signs with ordinary weak-field gravity, adopt the static isotropic metric

$$\begin{aligned} \mathrm{d}s^2 = &-\left(1+\frac{2\Phi}{c_0^2}\right)c_0^2\mathrm{d}t^2 \\ &+\left(1-\frac{2\Phi}{c_0^2}\right)\mathrm{d}\mathbf{x}^2, \end{aligned} \tag{13}$$

to first order in $|\Phi|/c_0^2$, with $\Phi \to 0$ at infinity. Here the relation between time and space factors is an additional physical assumption. Null propagation gives

$$n \simeq 1 - \frac{2\Phi}{c_0^2}, \qquad \Phi \simeq \frac{c_0^2}{2}(1-n). \tag{14}$$

Accordingly, added delays give a negative potential, whereas depletion gives a positive one relative to the exterior.

If clocks and massive probes couple to this same metric, stationary clocks obey $\mathrm{d}\tau/\mathrm{d}t \simeq 1 + \Phi/c_0^2$, and slow freely falling probes obey $\mathbf{a} = -\nabla\Phi$. For Eq. (7),

$$\Phi = \frac{c_0^2\varepsilon}{2}f, \qquad \mathbf{a} = -\frac{c_0^2\varepsilon}{2}\nabla f \tag{15}$$

at leading order. A central depletion with $f'(r) < 0$ is a potential hill and gives $a_r > 0$: outward free fall. Clocks there run faster relative to exterior clocks. Locally, ideal rods and clocks defined by the metric still measure the characteristic signal speed c_0.

The factor $1/2$ in Eq. (14) comes from the particular time–space relation in Eq. (13); it is not fixed by node counting. A different clock prescription can retain the same optical index and change the gravitational interpretation. Equations (13)–(15) specify a conditional completion of the model, not a derivation of Einstein's equations or of an equivalence principle.

V. A SMOOTH EXAMPLE

Consider a spherically symmetric encounter deficit

$$n(r) = 1 - \varepsilon e^{-r^2/L^2}, \qquad 0 < \varepsilon \ll 1. \tag{16}$$

It is regular at the origin and approaches the reference exponentially. The following observables illustrate the same depletion in two different optical experiments.

First, place endpoints at $(\mathbf{b}, -Z)$ and $(\mathbf{b}, Z)$, with $b = |\mathbf{b}|$ and Z fixed. The reference path is parallel to the z axis. Equation (8) gives

$$\Delta T = -\frac{\sqrt{\pi}\varepsilon L}{c_0}e^{-b^2/L^2}\,\mathrm{erf}(Z/L) + O(\varepsilon^2). \tag{17}$$

The coefficient of the advance approaches a finite limit for distant endpoints. This is a fixed-endpoint perturbation calculation; it does not compare arrival times of rays ending in different directions at infinity.

Second, send in a ray from the ordinary exterior with impact vector $\mathbf{b}$. Integrating Eq. (9) along its unperturbed trajectory gives the asymptotic change in direction

$$\Delta\mathbf{e}_\perp = 2\sqrt{\pi}\varepsilon\frac{\mathbf{b}}{L}e^{-b^2/L^2} + O(\varepsilon^2). \tag{18}$$

The deflection points away from the centre. A central ray has zero deflection by symmetry, and rays far outside the depleted region are unaffected to exponential accuracy.

Under the gravitational completion of Sec. IV,

$$\Phi(r) = \frac{c_0^2\varepsilon}{2}e^{-r^2/L^2}, \qquad a_r = \frac{c_0^2\varepsilon r}{L^2}e^{-r^2/L^2}. \tag{19}$$

These expressions are again leading-order relations. They give a regular outward acceleration for every $r > 0$.

One may additionally ask what source an observer using the Newtonian field equation would infer. Defining that source by

$$\nabla^2\Phi = 4\pi G\rho_{\mathrm{eff}} \tag{20}$$

gives, in three spatial dimensions,

$$\rho_{\mathrm{eff}}(r) = \frac{c_0^2\varepsilon}{8\pi GL^2}\left(\frac{4r^2}{L^2} - 6\right)e^{-r^2/L^2}. \tag{21}$$

The inferred source is negative in the centre and positive in an outer region. Its total integral vanishes at the order considered. The enclosed inferred mass is

$$M_{\mathrm{eff}}(r) = \frac{r^2\Phi'(r)}{G} = -\frac{c_0^2\varepsilon r^3}{GL^2}e^{-r^2/L^2}, \tag{22}$$

which is negative for finite $r > 0$ and tends to zero at infinity. Thus a local depleted profile can give outward acceleration without representing an isolated negative total mass. The fitted density ρ_{eff} is defined by Eq. (20); it is not the microscopic number density of removed nodes or a calculation of their energy.

VI. LONG-RANGE BEHAVIOR AND THE REMAINING PHYSICAL QUESTION

A negative-mass type exterior has a more specific requirement. Write $\eta = 1 - n$. If, outside a core,

$$\eta(r) \simeq \frac{\ell_*}{r}, \qquad \ell_* > 0, \tag{23}$$

then the weak-field completion gives

$$\Phi(r) \simeq \frac{c_0^2 \ell_*}{2r}, \qquad a_r \simeq \frac{c_0^2 \ell_*}{2r^2}. \tag{24}$$

This has the exterior form of a negative effective mass

$$M_{\rm eff} = -\frac{c_0^2 \ell_*}{2G}, \tag{25}$$

in the regime $r \gg \ell_*$. An exterior with exactly $n = 1$ has no such tail. For a smooth compactly supported deficit, the inferred total source in Eq. (20) is zero by the divergence theorem.

The long-range profile is additional physics. A finite set of removed nodes confined to a bounded region does not by itself establish Eq. (23). It would have to produce an extended change in the encounter density or in the effective processing rule of the remaining substrate. A microscopic theory would need to determine that response, its stability, and the dynamics of the objects used as clocks and probes. It would also have to establish that relevant signal species share the same continuum geometry. Frequency-dependent or species-dependent delays would give further observable structure.

The result established by the counting rule is narrower and direct: local contraction of processing sequences gives an optical index below the reference value, earlier traversal, and outward deflection for a central depleted profile. Applying the usual weak-field metric relation turns the same profile into faster interior clocks and repulsive free fall. The first set of statements follows from the continuum propagation model; the second identifies a precise additional hypothesis under which the missing-step intuition becomes a gravitational analogue.

[1] H. Kleinert and J. Zaanen, "Nematic world crystal model of gravity explaining absence of torsion in spacetime," Phys. Lett. A **324**, 361–365 (2004).

[2] T. Toffoli, "The Role of the Observer in Uniform Systems," in *Applied General Systems Research: Recent Developments and Trends*, edited by G. J. Klir (Plenum Press, New York, 1978), pp. 395–400, doi:10.1007/978-1-4757-0555-3_29.

[3] M. Visser, "Acoustic black holes: horizons, ergospheres and Hawking radiation," Class. Quantum Grav. **15**, 1767–1791 (1998).